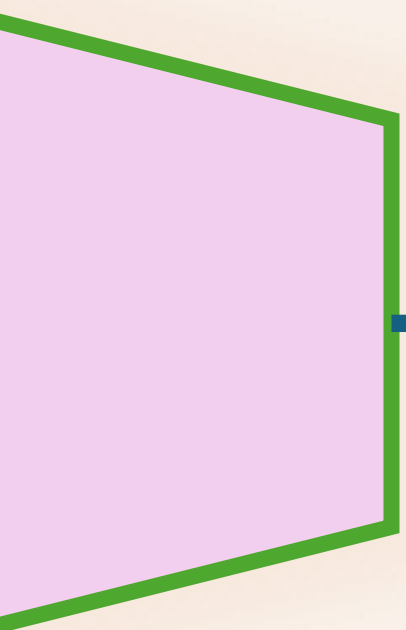


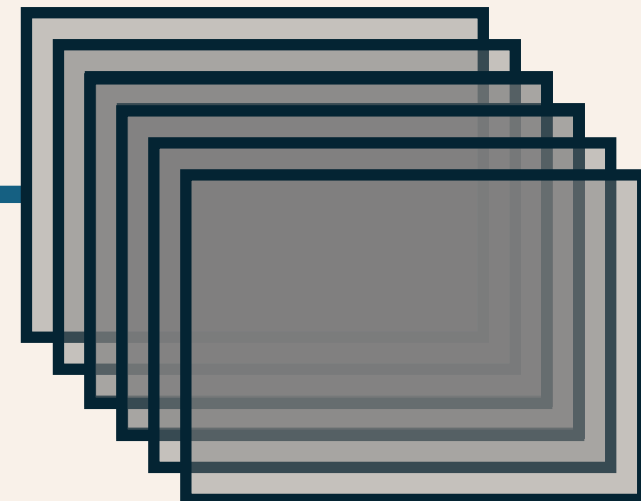
$I: B \times H \times W \times 3$

Input Image

Object Detection and ROI Warping



Feature Extraction



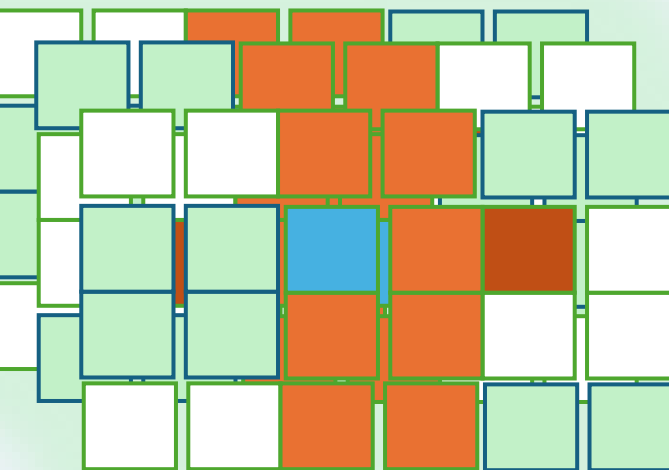
ROI Proposal
Generation

Multiple ROIs



ROI
Align

b, c



ROI Crop $A(b), K$

I_{roi}

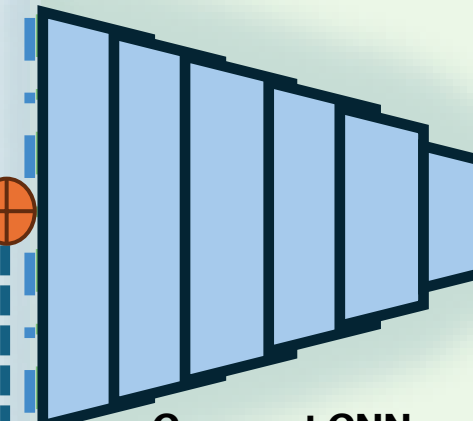
$[192 \times 192 \times 3]$

Fixed Edge Bank
(Sobel / Laplacian)

$[192 \times 192 \times 1]$

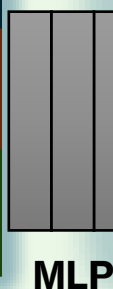
QRT-6D (pose head)

$FiLM(e_c \rightarrow \gamma, \beta)$



Compact CNN
encoder

Query bank/
Cross-attention



MLP

12×192

BiLSTM

$H_{rot}[192 \ 128, 6]$

$H_t[192 \ 128, 3]$

Output

Object Classes

Object BBox

Refined 6 DoF pose